StreamSense: Viewer Engagement & Retention Insights

University Capstone Project Report

Abstract

StreamSense is a viewer engagement analytics platform that helps streaming companies evaluate content using watch duration, completion percentage, pause frequency, and retention metrics instead of relying only on view counts.

Introduction

The project converts raw viewer session data into meaningful dashboards for acquisition teams to support content renewal and licensing decisions.

Problem Statement

Existing systems depend on total views and intuition, making it difficult to identify content that truly retains subscribers.

Objectives

Clean viewer data, calculate engagement scores, analyze retention, rank content, provide recommendations, and generate reports.

Technology Stack

Frontend: React.js, Backend: Node.js & Express.js, Database: MongoDB, Design: Figma, Authentication: JWT.

Modules

Login, Executive Dashboard, Viewer Analytics, Content Performance, Retention Analytics, Recommendations, Reports, and Settings.

Database Collections

Users, Subscription, Content, ViewerSession, EngagementScore, Retention, Recommendation, and Report.

Functional Requirements

Import data, clean missing values, calculate engagement scores, visualize analytics, filter data, and export reports.

Future Enhancements

Real-time analytics, AI-based recommendations, churn prediction, mobile application, and cloud deployment.

Conclusion

StreamSense enables data-driven content decisions by combining engagement and retention analytics into a simple, explainable dashboard.